

Predicting Superconducting Critical Temperature from Engineered Material Descriptors: A Comparative Machine Learning Study

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Abstract

Superconducting materials, which conduct electric current with zero resistance below a critical temperature (T_c), represent one of the most promising avenues for reducing energy losses in electrical infrastructure. Identifying material compositions associated with a higher T_c remains a central open problem in condensed matter physics and materials science. This study investigates whether machine learning models can predict T_c from a set of 81 engineered numerical descriptors derived from elemental and compositional properties. Six regression models — Linear Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, and a deep multilayer perceptron (MLP) — were trained and evaluated on a public superconductivity dataset (21,263 samples after cleaning) using an 80/20 train-test split and 5-fold cross-validation. Random Forest achieved the strongest performance (cross-validated RMSE ≈ 9.70 K, test $R^2 \approx 0.930$), narrowly ahead of XGBoost (CV-RMSE ≈ 9.88 K), while the deep MLP placed in the middle of the field (test $R^2 \approx 0.878$) and Linear Regression performed weakest ($R^2 \approx 0.738$), indicating a strongly nonlinear relationship between descriptors and T_c on which tree ensembles remain especially strong. Impurity-based and permutation feature importance agree that thermal-conductivity-, atomic-mass-, and valence-related descriptors are the most informative predictors, and Principal Component Analysis confirms that the dataset contains non-random structure related to T_c . We additionally address a data-leakage risk: the 21,263 measurements span only 15,542 distinct chemical formulas, so a random split shares compositions between train and test. Under a rigorous composition-grouped split (no chemical formula shared across partitions), Random Forest performance declines only modestly, from $R^2 \approx 0.930$ to $R^2 \approx 0.913$, and the model ranking is preserved — indicating that the models generalize to genuinely unseen compositions. We discuss the limitations of this proof-of-concept pipeline and outline directions for developing it into a screening tool for candidate superconducting materials.

Keywords: superconductivity; critical temperature; machine learning; random forest; XGBoost; deep learning; feature importance; materials informatics.

1. Introduction

1.1 Motivation

In 2026, severe flooding struck the Republic of Dagestan, Russia, and local officials attributed part of the resulting infrastructure damage to degraded electrical wiring — cables that wear out, overheat, and fail under stress. This observation motivated the present study: what if electrical grids did not lose energy to resistance at all?

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Superconductors are materials that conduct electric current with zero electrical resistance, meaning they do not dissipate energy as heat and, in principle, do not degrade through resistive overheating in the way conventional conductors do. Widespread use of superconducting materials in power infrastructure could substantially improve grid reliability and efficiency. The central obstacle to this vision is that most known superconductors only exhibit this property at extremely low temperatures, which makes them impractical for large-scale real-world deployment.

Identifying materials with a higher critical temperature (T_c) — the temperature below which superconductivity occurs — is therefore one of the most important open problems in modern condensed matter physics and materials science. This work applies machine learning methods to predict T_c from engineered material descriptors, as an initial step toward a computational screening pipeline for promising superconducting candidates.

1.2 Objectives

The specific objectives of this study are to: (i) compare the predictive performance of six regression algorithms — including a deep neural network — on a public superconductivity dataset; (ii) identify which engineered descriptors are most informative for predicting T_c ; and (iii) assess, through correlation analysis, dimensionality reduction, and a duplicate-leakage check, whether the dataset exhibits structure consistent with a learnable physical relationship and how robust the reported performance is.

2. Data and Methods

2.1 Dataset

The dataset used in this study consists of engineered numerical descriptors for superconducting materials, together with the corresponding critical temperature (`critical_temp`) as the target variable. Each descriptor is derived from elemental and compositional properties of the constituent atoms (e.g., atomic mass, valence, thermal conductivity, electron affinity, atomic radius, and fusion heat), aggregated using statistics such as mean, weighted mean, geometric mean, standard deviation, range, and entropy across the elements present in each material. The raw dataset contained 21,263 usable records after preprocessing, with 81 descriptor features. The data originate from the UCI Machine Learning Repository superconductivity dataset of Hamidieh [1].

2.2 Preprocessing

One column, `wtd_std_Density`, was coerced to a numeric type, and any rows with resulting invalid or missing values were removed. The target variable was set to `critical_temp`, and all remaining numerical columns were used as input features. The cleaned dataset was split into training (80%) and test (20%) subsets using a fixed random seed to ensure reproducibility.

2.3 Models

Six regression models were trained and compared:

- **Linear Regression**, as a baseline linear model.

- **Decision Tree Regressor**, a single nonlinear tree-based model.
- **Random Forest Regressor** (300 estimators), an ensemble of bagged decision trees.
- **Gradient Boosting Regressor**, a sequential boosting ensemble.
- **XGBoost Regressor** (300 estimators, max depth 6, learning rate 0.05, `subsample` 0.8, `colsample_bytree` 0.8), a regularized gradient boosting implementation.
- **Multilayer Perceptron (MLP)**, a deep feed-forward neural network implemented in PyTorch, with three hidden layers (256–128–64 units), batch normalization, ReLU activations, and dropout regularization. Inputs and target were standardized using training-set statistics, and the network was trained with the Adam optimizer under mean-squared-error loss with early stopping on an internal validation split.

All models were evaluated on the held-out test set using mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), and the coefficient of determination (R^2). To assess the stability of each model’s performance, 5-fold cross-validation was additionally performed on the training set, reporting the mean and standard deviation of the cross-validated RMSE (CV-RMSE, CV-STD).

2.4 Feature Importance and Structural Analysis

To interpret the best-performing model, impurity-based feature importances were extracted from the Random Forest model. Because impurity-based importances can be biased when features are correlated, permutation importance was additionally computed on the held-out test set as a robustness check. In parallel, Pearson correlation coefficients between each descriptor and `critical_temp` were computed across the full cleaned dataset. Finally, Principal Component Analysis (PCA) was applied to the standardized feature matrix to visualize the two-dimensional structure of the dataset with respect to critical temperature.

2.5 Duplicate and Leakage Analysis

Engineered superconductivity descriptors are known to contain many near-identical compositions. Because a random train–test split can place near-duplicate materials in both partitions and leak information, we performed two robustness checks. First, we counted exact-duplicate feature rows and re-evaluated the best model on a deduplicated dataset. Second, and more rigorously, we obtained the chemical formula of each material from the companion `unique_m.csv` file (which aligns one-to-one with the descriptor rows) and performed a *composition-grouped* split: all measurements sharing a chemical formula were kept entirely within either the training or the test partition, so that no formula appears in both. All six models were re-trained and re-evaluated under this leakage-free protocol using `GroupShuffleSplit` for the held-out test set and `GroupKFold` for 5-fold cross-validation, both keyed on the chemical formula.

3. Results

3.1 Model Comparison

Table 1 summarizes the performance of all six models on the test set and under 5-fold cross-validation. Random Forest achieved the best overall performance (test MAE = 5.08 K, RMSE = 8.97 K, $R^2 = 0.930$; CV-RMSE = 9.70 ± 0.25 K), closely followed by XGBoost (CV-RMSE = 9.88 ± 0.28 K); the two tree ensembles are statistically very close. The deep MLP placed in the middle of the field (test $R^2 = 0.878$, CV-RMSE = 12.11 ± 0.20 K), outperforming a single Decision Tree and Gradient Boosting on cross-validated RMSE but not matching the leading tree ensembles. Linear Regression performed markedly worse than all nonlinear models ($R^2 = 0.738$, CV-RMSE = 17.70 K).

Table 1: Performance comparison of six regression models on the test set, sorted by cross-validated RMSE.

Model	MAE	MSE	RMSE	R^2	CV-RMSE	CV-STD
Random Forest	5.076	80.452	8.970	0.930	9.700	0.253
XGBoost	5.811	86.037	9.276	0.925	9.876	0.280
MLP (Deep Learning)	7.795	140.725	11.863	0.878	12.114	0.201
Decision Tree	6.090	139.449	11.809	0.879	12.539	0.296
Gradient Boosting	8.501	152.150	12.335	0.868	12.794	0.149
Linear Regression	13.211	302.008	17.378	0.738	17.699	0.253

The gap between Linear Regression and the tree-based ensembles suggests that the relationship between the engineered descriptors and critical temperature is strongly nonlinear. Notably, the deep MLP — despite its capacity to model nonlinear interactions — did not surpass Random Forest or XGBoost, consistent with the broader observation that gradient-boosted and bagged tree ensembles remain strong baselines for tabular data with heterogeneous engineered features.

3.2 Feature Importance

Table 2 lists the descriptors with the highest impurity-based importance in the Random Forest model. A single feature, `range_ThermalConductivity`, dominates the importance ranking (0.536), followed by `wtd_gmean_ThermalConductivity` (0.126). The remaining top features relate to atomic mass, valence, and density, together accounting for a much smaller share of total importance.

Table 2: Top 8 features by Random Forest impurity-based importance.

Feature	Importance
<code>range_ThermalConductivity</code>	0.536
<code>wtd_gmean_ThermalConductivity</code>	0.126
<code>std_atomic_mass</code>	0.022
<code>wtd_gmean_Valence</code>	0.019
<code>std_Density</code>	0.011
<code>mean_Density</code>	0.011
<code>wtd_entropy_ThermalConductivity</code>	0.011
<code>gmean_ElectronAffinity</code>	0.011

We note that this pronounced concentration of importance in thermal-conductivity-related de-

scriptors may partly reflect multicollinearity among related features (e.g., `range_`, `wtd_gmean_`, and `wtd_std_ThermalConductivity` are derived from the same underlying property) rather than a single dominant physical mechanism; this is discussed further in Section 4.

3.3 Permutation Importance

To guard against the known bias of impurity-based importance under feature correlation, permutation importance was computed on the held-out test set (Table 3). The two measures agree closely: `range_ThermalConductivity` remains dominant (0.722 ± 0.019), followed by `wtd_gmean_ThermalConductivity`, `std_atomic_mass`, and `wtd_gmean_Valence`. The agreement between two independent importance measures strengthens confidence that these descriptors carry genuine predictive signal rather than being artifacts of a single scoring method.

Table 3: Top 8 features by Random Forest permutation importance (mean $\pm$ std over 5 repeats, held-out test set).

Feature	Permutation importance
<code>range_ThermalConductivity</code>	0.722 ± 0.019
<code>wtd_gmean_ThermalConductivity</code>	0.079 ± 0.005
<code>std_atomic_mass</code>	0.039 ± 0.002
<code>wtd_gmean_Valence</code>	0.033 ± 0.001
<code>mean_Density</code>	0.018 ± 0.001
<code>entropy_FusionHeat</code>	0.015 ± 0.001
<code>entropy_Density</code>	0.014 ± 0.000
<code>range_atomic_radius</code>	0.014 ± 0.001

3.4 Correlation Analysis

Pearson correlation coefficients between individual descriptors and `critical_temp` ranged from approximately -0.63 to $+0.72$. The strongest positive correlations were observed for `wtd_std_ThermalConductivity` ($r = 0.721$), `range_ThermalConductivity` ($r = 0.688$), `range_atomic_radius` ($r = 0.654$), and `std_ThermalConductivity` ($r = 0.654$). The strongest negative correlations were observed for `wtd_mean_Valence` ($r = -0.632$) and `wtd_gmean_Valence` ($r = -0.616$). These results are broadly consistent with the Random Forest feature importance ranking and indicate that thermal-conductivity-, radius-, and valence-related descriptors carry meaningful information about critical temperature, although correlation alone does not establish causal physical relationships.

3.5 Distribution and Structure of the Target Variable

The distribution of `critical_temp` across the dataset is right-skewed (skewness ≈ 0.86 ; mean ≈ 34.4 K, median ≈ 20.0 K, maximum 185 K): most materials exhibit relatively low critical temperatures, with a smaller number of materials reaching substantially higher values. This asymmetry means the RMSE is dominated by the harder-to-predict high- T_c tail and suggests that the dataset may contain multiple underlying material regimes with distinct superconducting behavior.

A two-dimensional PCA projection of the standardized feature matrix explained approximately 49.4% of total variance (PC1: 38.9%, PC2: 10.5%). Materials with higher critical temperatures tended to occupy specific regions of this projection rather than being uniformly distributed, indicating that the engineered descriptors capture structure related to critical temperature, although the moderate explained variance implies this two-dimensional view captures only part of the dataset’s complexity.

3.6 Duplicate and Leakage Robustness

A large fraction of the dataset consists of exact-duplicate feature vectors: 6,093 of 21,263 rows (28.7%) are exact duplicates of another row in feature space. Removing these duplicates leaves 15,170 unique rows. Retraining Random Forest on this deduplicated dataset lowers performance from $R^2 = 0.930$ (RMSE 8.97 K) to $R^2 = 0.911$ (RMSE 10.01 K).

The stronger check is the composition-grouped split. The 21,263 measurements correspond to only 15,542 distinct chemical formulas — some formulas (e.g., Y1Ba2Cu3O7) appear over one hundred times — so a random split shares many formulas between train and test. Under the leakage-free composition-grouped split (no formula shared across partitions; 4,274 test measurements), performance drops only modestly, confirming that the models genuinely generalize to unseen compositions rather than merely memorizing repeated ones. Table 4 reports all six models under both protocols. Random Forest, the best model, declines from $R^2 = 0.930$ to $R^2 = 0.913$ (RMSE 8.97 $\rightarrow$ 10.08 K, grouped CV-RMSE 9.94 K), and the overall model ranking is preserved. We regard the composition-grouped figures as the honest, leakage-controlled estimates of real-world screening performance.

Table 4: Random (leaky) versus composition-grouped (leakage-free) evaluation. Grouped columns use a formula-keyed `GroupShuffleSplit` test set and `GroupKFold` cross-validation; models sorted by grouped CV-RMSE.

Model	Test R^2		Test RMSE (K)		Grouped CV-RMSE
	Random	Grouped	Random	Grouped	
Random Forest	0.930	0.913	8.970	10.077	9.938
XGBoost	0.925	0.909	9.276	10.321	10.001
MLP (Deep Learning)	0.878	0.858	11.863	12.885	12.232
Gradient Boosting	0.868	0.854	12.335	13.049	12.796
Decision Tree	0.879	0.853	11.809	13.099	13.376
Linear Regression	0.738	0.726	17.378	17.877	17.639

Because this check removes leakage at the level of the chemical formula, the grouped figures should be regarded as the primary, honest estimate; the small size of the drop is itself an encouraging result.

4. Discussion and Limitations

The results indicate that engineered elemental descriptors carry substantial predictive signal for critical temperature, and that tree-based ensemble methods — particularly Random Forest and XGBoost — capture this signal considerably better than a linear model and, on this dataset,

better than a conventional deep neural network. However, several limitations should be considered when interpreting these results.

- The dataset uses engineered statistical descriptors rather than direct experimental or first-principles physical measurements, which limits the physical interpretability of the learned relationships.
- A large fraction of rows are exact-duplicate feature vectors (28.7%), and the 21,263 measurements span only 15,542 distinct chemical formulas, so a random split leaks similar compositions across train and test. We address this directly with a composition-grouped split (Section 3.6); the residual R^2 drop is small, but grouped performance ($R^2 \approx 0.913$ for Random Forest) should be read as the realistic estimate rather than the random-split figure.
- Feature importance and correlation analysis identify statistically predictive descriptors but do not establish causal physical mechanisms, particularly given the strong multicollinearity among thermal-conductivity-related features.
- Hyperparameters for the boosting methods and the neural network were only lightly tuned, so the relative ranking among the top models may shift under more systematic tuning.
- PCA is a linear dimensionality-reduction technique used here purely for visualization; it does not provide a direct physical interpretation of the principal components.

For these reasons, the present results should be regarded as a proof of concept rather than a definitive scientific conclusion about the physical determinants of superconducting critical temperature.

5. Future Work

- Testing the pipeline on additional, independently sourced superconductivity datasets.
- Using composition-grouped or deduplicated splitting to obtain leakage-free performance estimates.
- Combining engineered descriptors with more direct physical or first-principles measurements (e.g., density-functional-theory-derived features).
- Systematic hyperparameter tuning for all models, including the neural network architecture.
- Developing a ranking-based screening pipeline to prioritize candidate materials for experimental investigation.
- Adding uncertainty estimation (e.g., quantile regression or ensemble variance) to model predictions.

6. Conclusion

This study demonstrates that machine learning models can predict superconducting critical temperature from engineered material descriptors with meaningful accuracy. Among six compared models,

Random Forest achieved the best performance (cross-validated RMSE ≈ 9.70 K, $R^2 \approx 0.930$), narrowly outperforming XGBoost, while a deep multilayer perceptron placed in the middle of the field — confirming that gradient-boosted and bagged tree ensembles remain strong baselines for tabular materials data. The magnitude of improvement over the linear baseline indicates that the relationship between engineered descriptors and critical temperature is strongly nonlinear. Impurity-based and permutation feature importance, correlation analysis, and PCA visualization consistently point to thermal-conductivity-, valence-, mass-, and radius-related descriptors as informative predictors, though these findings should be interpreted as statistical associations rather than confirmed physical mechanisms. Crucially, a composition-grouped evaluation — in which no chemical formula is shared between training and test — shows that the models generalize to genuinely unseen compositions, with Random Forest declining only from $R^2 \approx 0.930$ to $R^2 \approx 0.913$; we regard this leakage-controlled figure as the honest estimate of real-world screening performance. Overall, this work serves as a proof-of-concept pipeline for machine-learning-assisted superconductor screening and provides a foundation for future extensions toward more physically grounded materials discovery workflows.

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